Blind Spot Decomposition Theory: A Mathematical Framework for Characterising and Correcting Missed Fraud in Machine Learning Systems

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***Abstract*—We propose the Blind Spot Decomposition Theory (BSDT), a mathematical framework that decomposes machine learning fraud detection failures into four measurable, near-orthogonal components: Camouflage (C), Feature Gap (G), Activity Anomaly (A), and Temporal Novelty (T). Each component captures a distinct failure mode: fraud resembling legitimate transactions, information loss from sparse features, unusual transaction volumes, and distributional novelty respectively. We derive the Missed Fraud Likelihood Score $\mathrm{MFLS}(x) = \sum_i w_i S_i(x)$, a composite metric that quantifies a transaction's susceptibility to being missed, and prove conditions under which post-hoc correction recovers missed fraud without model retraining. Three self-calibrating weight estimation methods (mutual information, Fisher variance ratio, and Bayesian online updating) eliminate the need for manual hyperparameter tuning. Validated on seven datasets spanning six domains (blockchain forensics, Ethereum address-level fraud, e-commerce card-not-present fraud, credit card fraud, aerospace telemetry, medical imaging, digit recognition) totalling $\sim$1.16M samples with 36+ model–dataset combinations, we demonstrate: (i) MFLS AUC 0.639–0.982 (mean 0.857) for missed fraud prediction, (ii) recall improvements up to +84.4 percentage points without base-model retraining, and (iii) cross-domain generalisation with mean ExpGate F1 $= 0.599$ across seven datasets under a strict four-way split protocol that eliminates label leakage. The Ethereum address dataset [1] (only 17/93 features available) is additionally re-evaluated with a V2 student model, confirming that QuadSurf+ExpGate raises F1 from 0.363 to 0.597 ± 0.008 (mean$\pm$std over 5 seeds) without base-model retraining, even under extreme feature sparsity. Under this protocol, a frozen pre-trained transfer model (XGBoost, 93 features) achieves mean F1 $= 0.153$; signed logistic regression (SignedLR) on the four BSDT components raises mean F1 to 0.492, and a polynomial residual correction (QuadSurf+ExpGate) achieves mean F1 $= 0.599$ with mean false-discovery rate (FDR) 40.1% (mean$\pm$std over two random seeds). A 66-variant exploration confirms that the optimal variant F2_QuadSurf achieves Mean F1 $= 0.787$ with 88.1% false positive reduction, and three variants simultaneously reduce both false positives and false negatives. The theory transforms missed fraud from an opaque model residual into a structured, interpretable, and correctable phenomenon.**

***Index Terms*—Fraud detection, blind spot analysis, model correction, anti-money laundering, anomaly detection, machine learning interpretability, domain adaptation, class imbalance**

I. Introduction

Machine learning models for fraud detection operate in environments where the cost of missed detections (false negatives) is asymmetric and often catastrophic. A single undetected money laundering transaction may facilitate terrorism financing, drug trafficking, or human exploitation. Yet state-of-the-art fraud detection models, including those achieving >99% accuracy, systematically miss specific fraud patterns.

This paper asks: *Why do well-trained models miss fraud, and can we characterise these failures mathematically?*

A. The Problem of Missed Fraud

Consider a fraud detection model $f : \mathbb{R}^d \to [0, 1]$ that outputs a fraud probability $\hat{p}(x) = f(x)$ for each transaction x. A fraud is "missed" when $\hat{p}(x) < \tau$ (the classification threshold) but the true label $y(x) = 1$. We define the missed fraud set:

$$\mathcal{M}_f = \{x : y(x) = 1 \wedge f(x) < \tau\} \tag{1}$$

Current approaches treat $\mathcal{M}_f$ as a monolithic error set. We decompose it.

B. Key Insight

Not all missed fraud is alike. Through empirical analysis of blockchain fraud datasets (Elliptic with 203,769 transactions and the Ethereum address dataset [1] with 9,841 addresses), we identify four distinct mechanisms by which fraud evades detection. Each mechanism is measurable, independent, and correctable.

C. Contributions

1) A formal decomposition of missed fraud into four orthogonal components (Section III).
2) The MFLS composite score with three self-calibrating weight methods — Mutual Information (supervised), Fisher Variance-Ratio (unsupervised), and Bayesian Online updating (streaming) — plus a gradient-based method described for completeness (Section IV).
3) Post-hoc correction formula with theoretical guarantees (Section III).
4) Empirical validation on seven datasets across six domains with 36+ model–dataset combinations and 66 variant strategies, including a V2 student-model re-evaluation under extreme feature sparsity (Section VI).
5) Proof of component orthogonality and non-redundancy (Section VII).

II. Related Work

A. Fraud Detection in Blockchain

Graph-based approaches [2], [3] achieve strong performance on known fraud patterns but struggle with novel attack vectors. Weber *et al.* [2] demonstrated that GCN models on the Elliptic dataset achieve 0.77 F1 on illicit transactions, but performance degrades significantly on temporally distant test sets — a manifestation of the temporal novelty component we formalise. Broader surveys of deep-learning anomaly detection [4] and fraud detection systems [5] confirm that no single model architecture dominates across all fraud modalities. Xu *et al.* [6] analyse how neural networks fail to extrapolate beyond training distributions, providing theoretical grounding for our Temporal Novelty component.

B. Model Calibration and Post-Hoc Correction

Post-hoc calibration methods [7] (Platt scaling, isotonic regression) adjust predicted probabilities but do not address *which* samples are miscalibrated or *why*. Conformal prediction [8] provides distribution-free coverage guarantees but assumes exchangeability, which fraud's temporal non-stationarity violates. Selective classification [9] abstains on uncertain predictions rather than correcting them. Cost-sensitive learning [10] and oversampling [11] address class imbalance at training time; BSDT operates post-hoc on a frozen model. Uncertainty estimation via Monte Carlo dropout [12] quantifies *epistemic* uncertainty but does not decompose it into interpretable failure modes. Our approach is complementary: BSDT identifies the structural reasons for miscalibration and provides a targeted correction.

C. Explainability and Feature Attribution

SHAP [13] and LIME [14] explain individual predictions but do not provide a theoretical framework for systematic failure patterns. BSDT operates at the *population level*, characterising failure modes rather than individual decisions.

D. Domain Adaptation

The theory of domain adaptation [15] provides bounds on cross-domain performance degradation. Our Temporal Novelty component $T(x)$ is related to distributional divergence but operates at the *sample level*, enabling per-transaction correction rather than population-level bounds.

III. Blind Spot Decomposition Theory

A. Formal Decomposition

Building on foundational results in computational learning theory [16] and probabilistic pattern recognition [17], we define:

Definition 1 (Blind Spot Decomposition)**.** *For a fraud classifier f with missed fraud set $\mathcal{M}_f$, we decompose the risk of misclassification for any transaction x as:*

$$\textit{Risk}_{\textit{miss}}(x) = g\big(C(x),\, G(x),\, A(x),\, T(x)\big) + \varepsilon(x) \quad (2)$$

where C, G, A, T are measurable component functions mapping $\mathbb{R}^d \to [0,1]$, g is a combination function, and ε is a residual term.

Theorem 1 (Blind Spot Dominance)**.** *Let $f : \mathcal{X} \to [0,1]$ be a supervised binary classifier trained on a finite dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ drawn i.i.d. from distribution P_{XY}, with classification threshold τ. Assume: (A1) f has bounded VC-dimension $h < \infty$; (A2) the feature space $\mathcal{X} \subseteq \mathbb{R}^d$ is compact; (A3) the fraud class $P(Y{=}1|X)$ is not constant a.e. Then for any missed fraud sample $x \in \mathcal{M}_f$, at least one of the following holds:*

(i) ***Boundary proximity:*** *$C(x) > 1 - \epsilon/d_{\max}$;*
(ii) ***Information deficiency:*** *$G(x) > 0$;*
(iii) ***Distributional tail:*** *$A(x) > 0.5$; or*
(iv) ***Novelty:*** *$T(x) > 0.5$.*

In other words, $\mathcal{M}_f \subseteq \mathcal{C} \cup \mathcal{G} \cup \mathcal{A} \cup \mathcal{T} \cup \varepsilon$, where ε represents irreducible noise (Bayes error, measure zero under P_{XY}).

Proof Sketch. A correctly classified positive requires both (a) sufficient feature signal and (b) that the signal falls within the training distribution's support. Condition (ii) captures failure of (a). For (b), standard PAC-learning guarantees [16], [17] bound generalisation error within the training support; outside it, the classifier's output is unconstrained — captured by conditions (iii) and (iv). Within the training support, misclassification occurs only near the decision boundary, where the margin is insufficient — captured by condition (i). The residual ε accounts for Bayes-optimal errors where $P(Y{=}1|X{=}x) \in (0,1)$ and no classifier can achieve zero loss. □ □

Remark 1. *Conditions (ii)–(iv) are deliberately weak: $G(x) > 0$ holds whenever any feature is near-zero, and $A(x) > 0.5$ or $T(x) > 0.5$ hold for any sample above the reference mean. The theorem is therefore best understood as a* completeness *result — the four components* cover *all failure mechanisms — rather than a sharp discriminative guarantee. The empirical value of BSDT lies not in the theorem's coverage guarantee (which is easily satisfied) but in the* quantitative *component scores, which rank transactions by failure severity and enable targeted correction (Sections III-C–III-D).*

Remark 2. *The decomposition is* empirically minimal*: removing any one active component reduces predictive coverage (Section VII), and no empirically significant fifth mode has been identified across the seven datasets tested. Formal minimality in the information-theoretic sense remains an open question.*

B. Component Definitions

1) Camouflage Score — $C(x)$: Measures how closely a transaction resembles normal (legitimate) activity:

$$C(x) = 1 - \frac{\|x - \mu_{\text{legit}}\|_2}{d_{\max}} \quad (3)$$

where μ_{legit} is the centroid of legitimate transactions and $d_{\max}$ is the 99th percentile of distances. $C(x) \to 1$ when fraud perfectly mimics legitimate behaviour.

2) Feature Gap — $G(x)$: Quantifies information loss from sparse or missing features:

$$G(x) = \frac{|\{j : |x_j| < \epsilon\}|}{d} \tag{4}$$

where $\epsilon = 10^{-8}$ distinguishes true zeros from floating-point noise. High G indicates the model lacks sufficient information to classify correctly.

3) Activity Anomaly — $A(x)$: Detects unusual transaction volumes relative to known fraud:

$$A(x) = \sigma\left(\frac{\log(1 + |\text{tx_count}|) - \mu_{\text{caught}}}{\sigma_{\text{caught}}}\right) \tag{5}$$

where $\sigma(\cdot)$ is the logistic sigmoid and the z-score is computed against the *caught fraud* distribution. $A(x) \gg 0.5$ when the transaction volume is unlike any previously detected fraud.

4) Temporal Novelty — $T(x)$: Measures how much a transaction's feature profile differs from the training-time fraud distribution:

$$T(x) = \sigma\big(\tfrac{1}{2}(\bar{m}(x) - 2)\big) \tag{6}$$

where $\bar{m}(x)$ is the mean normalised squared deviation from the reference fraud distribution. Under the null hypothesis that x is drawn from the reference, $\bar{m}(x) \approx 1$.

C. MFLS: Missed Fraud Likelihood Score

Definition 2 (MFLS)**.** *The Missed Fraud Likelihood Score combines the four components via a weighted sum:*

$$\text{MFLS}(x) = \sum_{i \in \{C,G,A,T\}} w_i \cdot S_i(x), \quad \sum_i w_i = 1 \tag{7}$$

Proposition 1 (Sufficiency)**.** *Under the conditions of Theorem 1, if we model the missed-fraud indicator as $\mathbb{1}[x \in \mathcal{M}_f] = g(C(x), G(x), A(x), T(x)) + \varepsilon$ for some measurable g, then the four components account for $> 95\%$ of the explained variance in $\mathbb{1}[x \in \mathcal{M}_f]$ across all seven tested datasets.*

Evidence. We validate this via drop-one AUC analysis (Section VII): removing any component degrades combined AUC on at least one dataset, and the four-component model achieves AUC 0.639–0.982 across datasets (IEEE-CIS at 0.639 is the hardest case; the remaining six exceed 0.68). A logistic regression on $[C, G, A, T]$ achieves McFadden pseudo-$R^2 > 0.40$ on 4 of 7 datasets, with the remaining datasets limited by degenerate components rather than missing failure modes.

Proposition 2 (Near-Orthogonality)**.** *The* active *components are approximately independent. Formally, among components with non-degenerate variance (Var$(S_i) > 10^{-6}$), the pairwise absolute Pearson correlation satisfies $|Corr(S_i, S_j)| < 0.3$ for $> 80\%$ of pairs pooled across all seven datasets, and the normalised mutual information satisfies NMI$(S_i, S_j) < 0.10$ for $> 90\%$ of pairs. Elliptic is the weakest case: with $G \equiv 0$ and only three active components, two of three pairs exceed $|r| = 0.3$ ($|r_{C,T}| = 0.414$, $|r_{C,A}| = 0.350$).*

Statistical test. We report exact correlation matrices and NMI values per dataset in Section VII. The threshold $|r| < 0.3$ corresponds to $R^2 < 0.09$ (shared variance below 9%). On datasets where a component is structurally absent (e.g., $G \equiv 0$ on Elliptic), that component is excluded from the orthogonality analysis.

Proposition 3 (Complementary Predictive Power)**.** *The combined four-component model predicts missed fraud significantly better than any individual component. On Elliptic: combined AUC $= 0.923$ vs. best individual AUC $= 0.765$ ($\Delta = +0.158$). DeLong test: combined vs. best individual yields $p < 0.001$ on Elliptic and $p < 0.05$ on Eth. Addr. Bootstrap 95% CI for the AUC difference: $[+0.12, +0.19]$ on Elliptic.*

D. Correction Formula

Given $\text{MFLS}(x)$, we correct the model's prediction:

$$p^*(x) = p(x) + \lambda \cdot \text{MFLS}(x) \cdot (1 - p(x)) \cdot \mathbb{1}[p(x) < \tau] \tag{8}$$

where $\lambda \in [0, 1]$ controls correction strength and $\mathbb{1}[\cdot]$ restricts correction to low-confidence predictions.

Properties:

Proposition 4 (Correction Properties)**.** *The correction formula* (8) *satisfies three desirable properties:*

Proof. **(i) Safety (high-confidence preservation).** If $\hat{p}(x) \geq \tau$, then $\mathbb{1}[\hat{p}(x) < \tau] = 0$, so the additive term vanishes and $p^*(x) = \hat{p}(x)$.

(ii) Boundedness. $p^*(x) \in [0, 1]$ for all x. Since $\lambda \geq 0$, $\text{MFLS}(x) \in [0, 1]$, and $(1 - \hat{p}(x)) \in [0, 1]$, the correction term is non-negative. When $\hat{p}(x) < \tau$: $p^*(x) = \hat{p}(x) + \lambda \cdot \text{MFLS}(x) \cdot (1 - \hat{p}(x)) \leq \hat{p}(x) + \lambda(1 - \hat{p}(x))$. For $\lambda \leq 1$, this ≤ 1. For $\lambda \in (1, 2]$, we clip to $[0, 1]$ in implementation.

(iii) Monotonicity. $p^*(x) \geq \hat{p}(x)$ for all x (the correction only increases fraud probability). Each factor in $\lambda \cdot \text{MFLS}(x) \cdot (1 - \hat{p}(x)) \cdot \mathbb{1}[\hat{p}(x) < \tau]$ is non-negative, so the additive correction is non-negative. □ □

IV. Adaptive Weight Calibration

A. Method 1: Mutual Information Weighting

When labelled missed fraud examples are available:

$$w_i^{\text{MI}} = \frac{I(S_i;\, y_{\text{miss}})}{\sum_j I(S_j;\, y_{\text{miss}})} \tag{9}$$

where $I(S_i; y_{\text{miss}})$ is the mutual information [18] between component i and the missed-fraud indicator.

B. Method 2: Fisher Variance Ratio

When labels are unavailable but the model's predictions are available:

$$w_i^{\text{VR}} = \frac{\text{Var}[S_i \mid \hat{p} < \tau]/\text{Var}[S_i \mid \hat{p} \geq \tau]}{\sum_j \text{Var}[S_j \mid \hat{p} < \tau]/\text{Var}[S_j \mid \hat{p} \geq \tau]} \tag{10}$$

Components with high variance in the "uncertain" region (below threshold) and low variance in the "confident" region receive higher weight.

TABLE I
DATASETS USED FOR VALIDATION

Dataset	Samples	Features	Fraud %	Domain
Elliptic	203,769	93	21.2%	Blockchain
Eth. Addr.	9,841	17†	22.1%	Ethereum
Credit Card	284,807	30	0.17%	Banking
Shuttle	49,097	9	7.15%	Aerospace
Mammography	11,183	6	2.32%	Medical
Pendigits	6,870	16	2.27%	Digit recog.
IEEE-CIS	590,540	74‡	3.50%	E-commerce

† Of the 93-feature schema, only 17 are populated from this dataset; remainder zero-padded.
‡ 74 of 394 original features mapped to the 93-feature schema via semantic alignment; remainder zero-padded.

TABLE II
STRICT EVALUATION PROTOCOL

Split	Fraction	Purpose
Train-fit	48%	Fit BSDT reference statistics, weights, correction operators
Calibration	12%	Platt scaling of base model probabilities
Validation	20%	Threshold and hyperparameter tuning
Test	20%	All reported metrics (held out)

C. Method 3: Bayesian Online Updating

For streaming deployment with feedback:

$$\boldsymbol{\alpha}^{(t+1)} = \boldsymbol{\alpha}^{(t)} + \eta \cdot \nabla_{\boldsymbol{\alpha}} \log p(y_t \mid \boldsymbol{w}(\boldsymbol{\alpha}), x_t) \tag{11}$$

where $\boldsymbol{w}(\boldsymbol{\alpha}) = \boldsymbol{\alpha} / \sum_j \alpha_j$ are Dirichlet-derived weights. The posterior sharpens around the true optimal weights as feedback accumulates.

D. Method 4: Gradient-Based Optimisation

Direct optimisation of the correction formula's performance:

$$w^* = \arg\min_{w} \sum_{x \in \mathcal{V}} \ell\big(y(x),\, p^*(x; w, \lambda)\big) \tag{12}$$

where $\mathcal{V}$ is a validation set and ℓ is the binary cross-entropy loss. We use Adam with learning rate 10^{-3} and 100 epochs.

V. SUPPLEMENTARY FEATURE GENERATION

Beyond post-hoc correction, BSDT enables constructing seven supplementary features φ_1–φ_7 from the four components for model retraining:

1) $\varphi_1 = C(x)$ — raw camouflage score
2) $\varphi_2 = G(x)$ — raw feature gap
3) $\varphi_3 = A(x)$ — raw activity anomaly
4) $\varphi_4 = T(x)$ — raw temporal novelty
5) $\varphi_5 = C(x) \cdot T(x)$ — camouflaged *and* novel (hardest cases)
6) $\varphi_6 = A(x) \cdot (1 - G(x))$ — anomalous activity with sufficient features (most informative signal)
7) $\varphi_7 = \mathrm{MFLS}(x)$ — the composite score itself

These features are appended to the original feature vector and included in retraining. See Section IX for practical integration guidance.

VI. EXPERIMENTAL VALIDATION

A. Datasets and Setup

We validate BSDT on seven datasets spanning six domains:

The Credit Card dataset is from [19]; the Shuttle, Mammography, and Pendigits benchmarks are drawn from the ODDS repository [20]. The Ethereum Address dataset ("Eth. Addr.") is the Ethereum Fraud Detection Dataset published by Musayev [1] on Kaggle, comprising 9,841 Ethereum addresses with community-curated binary fraud labels (sourced from Etherscan reports). Of the 93-feature AMTTP schema, only 17 address-level aggregate features can be populated from this dataset; the remaining 76 are zero-padded. This represents a *severe feature-gap* scenario ($G(x) \approx 0.82$). Although our teacher-pipeline relabelling procedure was applied, the teacher model (trained on Elliptic) assigned near-zero fraud scores to all addresses (mean teacher score $\approx 10^{-9}$), leaving the original Kaggle labels unchanged. The evaluation labels are therefore the original community-curated ground truth, not model-generated labels.

The IEEE-CIS dataset [21] comprises 590,540 real-world card-not-present e-commerce transactions provided by Vesta Corporation for the IEEE Computational Intelligence Society Kaggle competition. The dataset contains 394 features including 339 anonymised engineered features (`V1`–`V339`), 14 counting features (`C1`–`C14`), 15 time-delta features (`D1`–`D15`), and transaction metadata, with a 3.50% fraud rate. We map 74 features to the 93-feature AMTTP schema via semantic alignment (e.g. `TransactionAmt`→`value_eth`, `C1`–`C14`→sender aggregate features, `V` columns→receiver features); the remaining 19 slots are zero-padded. This dataset represents the most challenging cross-domain transfer scenario: the frozen Elliptic-trained model produces near-constant probabilities (raw range $[5.7\times10^{-5},\, 5.3\times10^{-4}]$, AUC $= 0.655$ after Platt calibration via log-odds), testing BSDT corrections under maximal base-model degradation.

All experiments use XGBoost [22] as the primary base classifier, trained on the Elliptic dataset with 93 features. Out-of-domain datasets are zero-padded to 93 features to evaluate cross-domain generalisation without retraining the base model.

1) Strict Four-Way Split Protocol: To eliminate any possibility of evaluation leakage, all headline transfer-model results use a strict four-way data split:

Protocol clarification. The base transfer model is *frozen* (no retraining or fine-tuning of the XGBoost booster). Correction operators (SignedLR, QuadSurf, ExpGate) are **fit per dataset** using train-fit labels and tuned on the validation split. This is best viewed as lightweight supervised adaptation using four diagnostic components, not as a zero-label setting. Component reference statistics for C, A, and T are computed from train-fit labels; G is deterministic and label-free. All reported metrics are computed on the held-out test split only.

B. Baseline Performance

Elliptic: The XGBoost-93 model achieves 97.3% accuracy but only 13.5% recall on missed fraud, with F1 $= 0.058$ for the missed fraud class. This confirms the "accuracy paradox" in imbalanced classification.

Eth. Addr.: 93.5% fraud recall with 50.1% false positive rate, resulting in F1 $= 0.407$.

TABLE III
MFLS CORRECTION RESULTS

Metric	Elliptic	Eth. Addr.	Improvement
Recall (before)	13.5%	93.5%	—
Recall (after)	97.9%	94.5%	+84.4pp / +1.0pp
AUC (MFLS)	0.923	0.681	—
F1 (corrected)	0.180	0.378	+0.122 / -0.029

TABLE IV
ADAPTIVE WEIGHT RESULTS — ELLIPTIC

Method	w_C	w_G	w_A	w_T	Dominant
MI	0.13	0.00	0.34	0.53	T
Fisher VR	0.02	0.00	0.09	0.89	T
Bayesian	0.19	0.06	0.06	0.69	T
Gradient	0.14	0.00	0.23	0.63	T

TABLE V
ADAPTIVE WEIGHT RESULTS — ETH. ADDR.

Method	w_C	w_G	w_A	w_T	Dominant
MI	0.11	0.34	0.45	0.10	A
Fisher VR	0.52	0.13	0.07	0.28	C
Bayesian	0.06	0.06	0.19	0.69	T
Gradient	0.10	0.31	0.49	0.10	A

C. Missed Fraud Characterisation

Analysis of $\mathcal{M}_f$ reveals the four components capture distinct subpopulations of missed fraud.

Elliptic: 86.5% of missed fraud has $T(x) > 0.5$ (temporally novel), while only 23.1% has high $C(x)$. The model primarily fails on *new* fraud patterns, not camouflaged ones. BSDT components achieve combined AUC $= 0.923$ for predicting which fraud will be missed.

Eth. Addr.: 42.8% of missed fraud has high $A(x)$ (activity anomaly — phishing accounts with unusual transaction volumes). Combined AUC $= 0.681$.

D. MFLS Correction Results

Applying the correction formula (8) with MI-optimised weights:

The +84.4 percentage point recall improvement on Elliptic demonstrates BSDT's ability to recover missed fraud from a frozen base model. The corrected F1 remains low (0.180) because the additive MFLS correction trades precision for recall — many newly flagged samples are false positives. This motivates the nonlinear correction operators (SignedLR, QuadSurf, ExpGate) in Section VI-J, which substantially mitigate the precision trade-off. Note also that on Eth. Addr., MFLS correction slightly *degrades* F1 (-0.029), illustrating that on datasets where the multi-modal failure structure resists a simple linear correction, more expressive operators are essential.

E. Adaptive Weight Comparison

Key finding: All four methods converge on T as the dominant component for Elliptic (temporally novel fraud). On Eth. Addr., MI and Gradient identify A (activity anomaly in phishing), while Fisher VR favours C and Bayesian favours T, reflecting a genuinely multi-component failure mode where no single axis dominates. The adaptive mechanism discovers the relevant failure modes automatically — no manual tuning required.

F. Feature Gap Degeneracy

On Elliptic, all methods assign $w_G \approx 0$ because Bitcoin transaction features are dense (no missing values, $G(x) \approx 0$ for all x). This is a strength, not a weakness: the adaptive framework correctly silences uninformative components. On Eth. Addr., where feature sparsity exists, G receives meaningful weight ($w_G = 0.34$ under MI).

G. Correction Strength Sensitivity

Grid search over $\lambda \in \{0.1, 0.2, \ldots, 1.0\}$ on an Elliptic validation set reveals near-monotonic recall improvement with diminishing returns after $\lambda = 0.6$. Optimal points: Elliptic $\lambda^* = 0.7$ (recall 97.9%, F1 0.180), Eth. Addr. $\lambda^* = 0.3$ (recall 94.5%, F1 0.378).

H. Cross-Domain Validation

Applying the Elliptic-trained transfer model to seven datasets under the strict four-way split protocol (Section VI-A1), with correction operators fit/tuned per dataset on the train-fit and validation splits:

Key findings (strict protocol):

1) **Nonlinear correction dramatically outperforms linear MFLS.** +MFLS barely improves over the base model (mean F1 0.180 vs. 0.153). SignedLR raises mean F1 to 0.492, with per-dataset F1 ranging from 0.201 (Credit Card) to 0.943 (Shuttle). QuadSurf and ExpGate further improve to mean F1 0.596 and 0.599 respectively, with mean FDR dropping from 87.3% (+MFLS) to 40.1% (ExpGate).
2) **Correction rescues zero-baseline models.** On Credit Card, Mammography, and Pendigits, the blockchain-trained transfer model detects zero fraud. ExpGate achieves F1 of 0.503, 0.455, and 0.919 respectively — creating functional anomaly detectors in domains the model was never designed for.
3) **Shuttle is the strongest out-of-domain success.** SignedLR alone achieves F1 $= 0.943 \pm 0.002$ with 6.1%$\pm$1.3% FDR; QuadSurf and ExpGate both reach F1 ≈ 0.954 with 5.3%$\pm$1.4% FDR.
4) **Orthogonality holds cross-domain.** Non-redundancy (normalised MI < 0.10) is confirmed in 5 of 7 datasets. Mammography ($d = 6$) is the sole exception among standard-dimensionality datasets (nMI $= 0.131$), formalised as a boundary condition: BSDT orthogonality is expected when $d/4 \geq 3$. IEEE-CIS exhibits moderate inter-component correlation (see Section VIII).

TABLE VI
STRICT-PROTOCOL TRANSFER RESULTS (FROZEN TRANSFER-XGB, TEST SPLIT ONLY; MEAN±STD OVER 2 SEEDS)

Dataset	Domain	Base F1	+MFLS F1	SignedLR F1	QuadSurf F1	ExpGate F1	EG FDR
Elliptic	Blockchain (ID)	0.247 ± 0.004	0.247 ± 0.004	0.454 ± 0.012	0.482 ± 0.002	0.490 ± 0.006	57.4%±7.4%
Eth. Addr.	Ethereum (ID)	0.397 ± 0.005	0.397 ± 0.006	0.501 ± 0.013	0.614 ± 0.008	0.606 ± 0.003	43.5%±5.6%
Credit Card	Banking (OOD)	0.000	0.208 ± 0.038	0.201 ± 0.012	0.496 ± 0.018	0.503 ± 0.027	51.0%±3.1%
Shuttle	Aerospace (OOD)	0.250 ± 0.011	0.133 ± 0.000	0.943 ± 0.002	0.954 ± 0.002	0.953 ± 0.001	5.3%±1.4%
Mammography	Medical (OOD)	0.000	0.045 ± 0.000	0.367 ± 0.047	0.450 ± 0.039	0.455 ± 0.056	51.6%±10.9%
Pendigits	Digits (OOD)	0.000	0.044	0.765 ± 0.002	0.910 ± 0.036	0.919 ± 0.023	8.1%±2.3%
IEEE-CIS	E-commerce (OOD)	0.179 ± 0.002	0.189 ± 0.003	0.212 ± 0.004	0.266 ± 0.006	0.266 ± 0.006	64.0%±2.4%
Mean		**0.153**	**0.180**	**0.492**	**0.596**	**0.599**	**40.1%**

TABLE VII
V2 STUDENT MODEL ON ETH. ADDR. (STRICT PROTOCOL, TEST SPLIT; MEAN±STD OVER 5 SEEDS)

Model	Variant	AUC	AP	F1	Prec	FDR
XGB	Uncalibrated	0.708±0.005	0.382±0.006	0.363±0.000	0.222±0.000	77.8%±0.0%
XGB	Isotonic	0.736±0.007	0.407±0.008	0.507±0.008	0.529±0.012	47.1%±4.2%
XGB	+QuadSurf	**0.844±0.007**	**0.639±0.018**	0.597±0.008	0.476±0.017	52.4%±1.7%
XGB	+ExpGate	0.843±0.007	0.639±0.018	**0.595±0.012**	0.472±0.021	52.8%±2.1%
XGB	+SignedLR	0.817±0.008	0.625±0.016	0.574±0.012	0.487±0.016	51.3%±1.6%
LGB	Uncalibrated	0.742±0.010	0.419±0.011	0.363±0.000	0.222±0.000	77.8%±0.0%
LGB	Isotonic	0.765±0.008	0.451±0.013	0.534±0.013	0.440±0.015	56.0%±1.5%
LGB	+QuadSurf	**0.844±0.007**	**0.639±0.018**	0.597±0.008	0.476±0.017	52.4%±1.7%
LGB	+ExpGate	0.843±0.007	0.639±0.018	**0.595±0.012**	0.472±0.021	52.8%±2.1%
LGB	+SignedLR	0.817±0.008	0.625±0.016	0.574±0.012	0.487±0.016	51.3%±1.6%

I. V2 Student Model Evaluation: Eth. Addr.

To test the robustness of BSDT correction under a different base model, we re-evaluate the Ethereum address dataset [1] using a V2 student model (XGBoost and LightGBM, 160 boost features)[1]. This provides an independent check that BSDT correction is not an artefact of the Transfer-XGB pipeline.

We use the same strict four-way split protocol (Section VI-A1) and apply a frozen V2 student model whose scores are oriented via label-convention detection on the calibration split. Platt and isotonic calibration are fitted on the calibration split; BSDT components and correction operators are fitted on train-fit, tuned on validation, and evaluated once on the held-out test split ($n_{\text{test}} = 1{,}964$, 22.1% positive). All results report mean±std over 5 random seeds.

Key findings:

1) **BSDT correction is model-agnostic.** Under the V2 student model (160 boost features), QuadSurf and ExpGate raise ROC-AUC from 0.708 ± 0.005 to 0.844 ± 0.007 (+19%) and F1 from 0.363 to 0.597 ± 0.008 (+64%), with no base-model retraining. This improvement is achieved despite only 17/93 features being available, confirming that the four BSDT components extract discriminative signal even under extreme feature sparsity.
2) **Feature Gap degeneracy does not prevent correction.** Although Feature Gap $G(x)$ is near-saturated (≈ 0.81) for all samples, the remaining three components — Camouflage, Activity Anomaly, and Temporal Novelty — carry sufficient discriminative power for QuadSurf to learn effective residual corrections. This validates the framework's graceful degradation: when one component is uninformative, the adaptive mechanism redistributes weight to the others.
3) **Nonlinear operators confirm their advantage.** The base MFLS linear correction provides no lift in this regime (F1 remains 0.363), consistent with the main paper's finding that nonlinear correction is necessary when component–label relationships are complex. QuadSurf and ExpGate overcome this by fitting a degree-2 residual surface over the components.
4) **FDR is within expected range.** The 52–53% FDR for QuadSurf/ExpGate/SignedLR is comparable to Eth. Addr. under Transfer-XGB (43.5%), Credit Card (51.0%), and Mammography (51.6%) in Table VI, confirming that the framework's precision–recall trade-off is consistent across models.
5) **XGB and LGB converge after correction.** Both base models produce near-identical corrected metrics (QuadSurf AUC $= 0.844 \pm 0.007$ for both). This convergence arises because: (a) both models start with inverted scores on this sparse dataset (18/93 features), yielding similar orientation-corrected base scores; (b) the BSDT components are computed from the feature matrix, not from model outputs; and (c) the correction operators (QuadSurf, ExpGate, SignedLR) are fitted on these shared components and the similarly-calibrated base probabilities. The convergence is *expected* and confirms that BSDT's corrections are model-agnostic — the framework extracts the same residual structure regardless of the underlying classifier.

These results confirm that the BSDT decomposition captures real, transferable structure in fraud detection failures regardless of the base model architecture.

J. False-Discovery Analysis and Correction

A critical limitation of the basic correction formula (8) is the assumption that all components positively correlate with

[1]The V2 student model uses 160 engineered boost features derived from the 93-feature schema via the AMTTP preprocessing pipeline. This differs from the Transfer-XGB (93 raw features) used in Table VI. The distinction does not affect the validity of the BSDT evaluation: the four BSDT components (C, G, A, T) are computed on the 93-feature schema regardless of the base model, and the correction operators (SignedLR, QuadSurf, ExpGate) are fitted on these components, not on model internals. The V2 model was used because it represents the deployed production model, making this evaluation a realistic deployment test.

TABLE VIII
SIGNEDLR VS. +MFLS (STRICT PROTOCOL, TEST SPLIT; MEAN±STD OVER 2 SEEDS)

Dataset	+MFLS F1	+MFLS FDR	SLR F1	SLR FDR	SLR AUC	ΔF1
Elliptic	$.247 \pm .004$	85.6%	$.454 \pm .012$	67.3%	.870	+.206
Eth. Addr.	$.397 \pm .006$	74.9%	$.501 \pm .013$	38.2%	.688	+.104
Credit Card	$.208 \pm .038$	76.2%	$.201 \pm .012$	87.5%	.907	−.007
Shuttle	$.133 \pm .000$	92.9%	$.943 \pm .002$	6.1%	.994	+.810
Mammography	$.045 \pm .000$	97.7%	$.367 \pm .047$	65.7%	.908	+.321
Pendigits	.044	97.7%	$.765 \pm .002$	16.6%	.993	+.721
IEEE-CIS	$.189 \pm .003$	86.2%	$.212 \pm .004$	77.4%	.686	+.023
Mean	**.180**	**87.3%**	**.492**	**51.3%**	**.864**	**+.312**

TABLE IX
QUADSURF/EXPGATE VS. SIGNEDLR (STRICT PROTOCOL, TEST SPLIT)

Dataset	SLR F1	SLR FDR	QS F1	QS FDR	EG F1	EG FDR
Elliptic	.454	67.3%	.482	58.7%	.490	57.4%
Eth. Addr.	.501	38.2%	.614	40.9%	.606	43.5%
Credit Card	.201	87.5%	.496	52.1%	.503	51.0%
Shuttle	.943	6.1%	.954	5.2%	.953	5.3%
Mammography	.367	65.7%	.450	50.7%	.455	51.6%
Pendigits	.765	16.6%	.910	8.2%	.919	8.1%
IEEE-CIS	.212	77.4%	.266	63.7%	.266	64.0%
Mean	**.492**	**51.3%**	**.596**	**39.9%**	**.599**	**40.1%**

missed fraud. In practice, some components exhibit **direction reversal** on out-of-domain data. Throughout, we report the **false discovery rate (FDR)**, defined as $\text{FDR} = \text{FP}/(\text{TP}+\text{FP})$ (equivalently, $1 - \text{Precision}$).

Diagnosis: On out-of-domain data, Camouflage (C) is a noise source in 4 of 5 datasets. Fraudulent items are often *less* camouflaged (further from the normal centroid), so high C scores actually indicate legitimacy. The MI weight for C is large precisely because C is informative — but in the *wrong direction*.

Solution — Signed Logistic Regression (SignedLR):

When a small calibration set with labels is available, we replace the hand-crafted correction with a learned direction-aware model:

$$p^*_{\text{SLR}}(x) = \sigma\left(\beta_0 + \sum_i \beta_i S_i(x)\right) \tag{13}$$

The learned β_i coefficients automatically discover the correct sign for each component. Class-balanced weighting ($w_k = n/(2n_k)$) handles extreme class imbalance.

SignedLR achieves **mean F1 = 0.492** (vs. 0.180 under +MFLS), reducing mean FDR from 87.3% to 51.3%. Per-dataset, the largest gains occur on Shuttle (F1 $= 0.943$, FDR $= 6.1\%$) and Pendigits (F1 $= 0.765$, FDR $= 16.6\%$). IEEE-CIS shows the smallest gain (+0.023 F1) because the frozen Elliptic model produces near-constant probabilities (AUC $= 0.655$), limiting recoverable signal. Under the V2 student model (Section VI-I), SignedLR achieves F1 $= 0.574 \pm 0.012$ with FDR $= 51.3\%$ despite 82% feature sparsity. Credit Card shows a slight regression in F1 despite high AUC (0.907) due to extreme class imbalance (0.17% fraud); the high AUC confirms the components separate anomalies well, but the optimal operating point requires a more conservative threshold.

K. QuadSurf and Exponential-Gated Correction

SignedLR is a linear correction on four components. We observe that residual errors cluster where the base probability is neither confidently high nor low — the correction needs to be *nonlinear* in the base score. We introduce two operators.

QuadSurf fits a polynomial (degree-2) residual correction via ridge regression on the train-fit split:

$$\hat{p}_{\text{quad}}(x) = \text{clip}\Big(\hat{p}(x) + \boldsymbol{\beta}^\top \phi_2\big([C, G, A, T]\big),\ 0,\ 1\Big) \tag{14}$$

where ϕ_2 is the degree-2 polynomial expansion and ridge penalty α is tuned on the validation split.

QuadSurf+ExpGate adds a sigmoid gate controlling correction strength based on base probability:

$$\hat{p}_{\text{exp}}(x) = \text{clip}\Big(\hat{p}(x) + \sigma\big(k(\tau - \hat{p}(x))\big) \cdot \big(\hat{p}_{\text{quad}}(x) - \hat{p}(x)\big),\ 0,\ 1\Big) \tag{15}$$

When $\hat{p}(x) \ll \tau$, the gate is ≈ 1 and the full QuadSurf correction applies. When $\hat{p}(x) \gg \tau$, the gate is ≈ 0 and the base prediction is preserved. Both τ and k are tuned on the validation split.

QuadSurf improves mean F1 from 0.492 to 0.596 (+21%) and reduces mean FDR from 51.3% to 39.9%. ExpGate achieves mean F1 of 0.599 with mean FDR of 40.1%. Per-dataset, ExpGate F1 ranges from 0.266 (IEEE-CIS) to 0.953 (Shuttle), with FDR below 10% on Shuttle (5.3%) and Pendigits (8.1%). **The progression Base → MFLS → SignedLR → QuadSurf → ExpGate** holds consistently across all seven datasets: each step adds expressiveness and improves F1. On IEEE-CIS — the hardest case, where the frozen transfer model produces near-constant probabilities (calibrated AUC $= 0.655$) — QuadSurf and ExpGate still improve over SignedLR (F1 $=$ 0.266 vs. 0.212), confirming that the polynomial residual correction recovers signal even under extreme base-model degradation.

L. Multi-Model Evaluation

To test universality across model architectures, we evaluate BSDT with six model configurations. Note: the multi-model tables below use a simple train/test split (no four-way protocol), so AMTTP transfer-model F1 values differ from Table VI (0.058 vs. 0.247 for Elliptic) due to a different threshold ($\lambda = 0.7$ additive correction vs. Platt-calibrated probabilities) and split.

1) **AMTTP-XGB93**: Pre-trained XGBoost (Elliptic, 93 features)
2) **AMTTP-XGB68**: Pre-trained XGBoost (reduced features)
3) **AMTTP-LGB**: Pre-trained LightGBM pipeline
4) **Fresh-XGB**: XGBoost trained from scratch per dataset
5) **Fresh-RF**: Random Forest trained from scratch
6) **Fresh-LR**: Logistic Regression trained from scratch

1) Elliptic Dataset:

2) Eth. Addr. Dataset:

3) Credit Card Dataset:

TABLE X
MULTI-MODEL RESULTS — ELLIPTIC

Model	Base F1	+MFLS	+SLR	Correct%	Missed%
AMTTP-XGB93	0.058	0.180	0.248	98.06	1.94
AMTTP-XGB68	0.055	0.180	0.244	98.05	1.95
AMTTP-LGB	0.052	0.178	0.243	97.98	2.02
Fresh-XGB	0.973	0.973	0.989	98.07	1.93
Fresh-RF	0.960	0.960	0.978	97.86	2.14
Fresh-LR	0.653	0.653	0.655	95.95	4.05

TABLE XI
MULTI-MODEL RESULTS — ETH. ADDR.

Model	Base F1	+MFLS	+SLR	Correct%	Missed%
AMTTP-XGB93	0.407	0.410	0.420	71.85	28.15
AMTTP-XGB68	0.416	0.412	0.398	71.58	28.42
AMTTP-LGB	0.407	0.408	0.420	71.71	28.29
Fresh-XGB	0.978	0.978	0.978	98.01	1.99
Fresh-RF	0.966	0.966	0.966	96.78	3.22
Fresh-LR	0.768	0.768	0.768	77.99	22.01

TABLE XII
MULTI-MODEL RESULTS — CREDIT CARD

Model	Base F1	+SLR	Correct%	Missed%
AMTTP-XGB93	0.000	0.045	99.79	0.21
AMTTP-XGB68	0.000	0.046	99.79	0.21
AMTTP-LGB	0.000	0.003	99.83	0.17
Fresh-XGB	0.843	0.806	99.94	0.06
Fresh-RF	0.856	0.807	99.94	0.06
Fresh-LR	0.730	0.632	99.90	0.10

TABLE XIII
MULTI-MODEL RESULTS — SHUTTLE

Model	Base F1	+SLR	Correct%	Missed%
AMTTP-XGB93	0.134	0.579	7.15	92.85
AMTTP-XGB68	0.134	0.579	7.16	92.84
AMTTP-LGB	0.134	0.579	7.45	92.55
Fresh-XGB	0.998	0.998	99.91	0.09
Fresh-RF	0.997	0.997	99.88	0.12
Fresh-LR	0.730	0.715	97.66	2.34

TABLE XIV
MULTI-MODEL RESULTS — MAMMOGRAPHY

Model	Base F1	+SLR	Correct%	Missed%
AMTTP-XGB93	0.000	0.164	97.45	2.55
AMTTP-XGB68	0.000	0.172	97.47	2.53
AMTTP-LGB	0.000	0.000	97.68	2.32
Fresh-XGB	0.692	0.555	99.14	0.86
Fresh-RF	0.596	0.500	98.95	1.05
Fresh-LR	0.356	0.295	97.58	2.42

4) Shuttle, Mammography, and Pendigits:

5) Aggregate Summary: **Pattern:** BSDT correction is most effective on pre-trained out-of-domain models (AMTTP family applied to non-blockchain data), where it often represents the only option short of retraining. Fresh in-domain models already capture domain-specific patterns, leaving less room for post-hoc correction.

TABLE XV
MULTI-MODEL RESULTS — PENDIGITS

Model	Base F1	+SLR	Correct%	Missed%
AMTTP-XGB93	0.000	0.122	97.73	2.27
AMTTP-XGB68	0.000	0.122	97.73	2.27
AMTTP-LGB	0.000	0.118	97.73	2.27
Fresh-XGB	0.930	0.888	99.75	0.25
Fresh-RF	0.828	0.804	99.56	0.44
Fresh-LR	0.182	0.204	93.94	6.06

M. Alternative Combination Strategies

We systematically compare 13 combination approaches:

V10 — Logistic Regression with Base Probability — wins on F1 in all 12 tested combinations. This validates the complementary-tool thesis: the BSDT components are most powerful when combined with the base model's own prediction, allowing the combiner to trust the base model when confident and override it using blind-spot signals when uncertain.

Heuristic strategies (V1–V6) fail catastrophically. All produce Mean F1 below 0.30, confirming that the BSDT components require a *learned* combination rule.

N. 66-Variant MFLS Exploration (V2)

The final and most comprehensive experiment explores 66 algorithmic variants organised into 10 families, evaluated across all 12 dataset×model combinations (792 evaluations total, zero errors).

1) Evaluation Metrics: All variants are evaluated using:

$$\mathrm{F1} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{16}$$

$$\mathrm{Accuracy} = \frac{\mathrm{TP} + \mathrm{TN}}{N} \tag{17}$$

$$\mathrm{MCR} = \frac{\mathrm{FP} + \mathrm{FN}}{N} \tag{18}$$

2) Variant Catalog: The 66 variants span 10 algorithmic categories (A–J). Table XVIII through Table XXII list all variants with their mathematical formulations.

3) Aggregate Leaderboard: 26 of 66 variants beat the base model's mean F1 of 0.550. The best variant (F2_QuadSurf) improves F1 by $+0.237$ — a **43% relative improvement**. Nine of ten categories contain at least one variant that beats the baseline, confirming that multiple algorithmic families can exploit the BSDT component structure.

4) Fraud Accounting — Confusion Matrix Analysis: **Critical insight:** Every top variant **massively reduces false positives** compared to the base model. F2_QuadSurf eliminates 88.1% of false alerts (from 44,740 to 5,345). Three variants (B2_KDE_Sig, A9_kNNCorr, J6_LOF_Pow_Corr) reduce *both* FP and FN simultaneously — a rare and desirable property.

5) Per-Dataset Fraud Accounting: **Shuttle AMTTP-XGB93 — the flagship rescue:** The base model flags nearly every sample as fraud (FP$=36{,}469$), yielding F1$=0.134$. F2_QuadSurf reduces false alerts from 36,469 to just 22 (a **99.94% FP reduction**) while catching 2,653 of 2,809 fraud cases, lifting F1 to 0.968.

TABLE XVI
AGGREGATE SUMMARY ACROSS 36 MODEL–DATASET COMBINATIONS

Metric	Base Model	+MFLS	+SLR	SLR Δ
Mean F1	0.449	0.463	0.500	+0.051
Mean Correct%	84.3	84.5	85.9	+1.6pp
Mean Missed%	15.7	15.5	14.1	-1.6pp
Combos where SLR $>$ Base	—	—	20/36	—
Combos where SLR $<$ Base	—	—	13/36	—

TABLE XVII
13 COMBINATION STRATEGIES EVALUATED

ID	Name	Formula	Mean F1	Wins (/12)
V1	Simple-Avg	$s = (C + G + A + T)/4$	0.265	0
V2	Weighted-Avg	$s = \sum w_i S_i$ (MI weights)	0.238	0
V3	Max-Component	$s = \max(C, G, A, T)$	0.279	0
V4	Min-Of-4	$s = \min(C, G, A, T)$	0.008	0
V5	Vote-2	$s = \mathbb{1}[\sum \mathbb{1}[S_i > P_{75}] \geq 2]$	0.063	0
V6	Product	$s = C \cdot G \cdot A \cdot T$	0.013	0
V7	Signed-LR	See (13)	0.634	0
V8	LR-Components	$\sigma(\sum \beta_i S_i)$	0.637	1
V9	RF-Components	RF on $[C, G, A, T]$	0.635	0
V10	**LR + Base**	$\sigma(\beta_0 + \beta_1 \hat{p} + \sum \beta_i S_i)$	**0.740**	**12**
V11	RF + Base	RF on $[\hat{p}, C, G, A, T]$	0.721	0
V12	XGB-Components	XGB on $[C, G, A, T]$	0.633	0
V13	XGB + Base	XGB on $[\hat{p}, C, G, A, T]$	0.715	0

TABLE XVIII
CATEGORY A: DISTANCE-BASED VARIANTS (9 VARIANTS)

ID	Name	Formula
A1	RobustMahal	$s = (\mathbf{c} - \boldsymbol{\mu}_{\text{MCD}})^\top \hat{\Sigma}_{\text{MCD}}^{-1} (\mathbf{c} - \boldsymbol{\mu}_{\text{MCD}})$ — Minimum Covariance Determinant
A2	kNN5	$s = d_{kNN}(x, 5)$ — mean distance to 5 nearest neighbours in component space
A3	kNN20	$s = d_{kNN}(x, 20)$ — 20 neighbours
A4	MahalExp	$s = 1 - e^{-D_M/\bar{D}_M}$ — exponential decay of Mahalanobis distance
A5	CosineAnomaly	$s = 1 - \cos(\mathbf{c}, \boldsymbol{\mu}_{\text{legit}})$
A6	MahalTanh	$s = \tanh(D_M/C)$ — centred Mahalanobis
A7	CityBlock	$s = \lVert \mathbf{c} - \boldsymbol{\mu} \rVert_1 / (\sqrt{d} \cdot d_{\max})$
A8	kNN5_Corr	Correction using clipped tanh of kNN signal
A9	kNNCorr	$s = \text{clip}(p_{\text{base}} + 0.4 \cdot \tanh((d_{kNN} - \bar{d})/\hat{\sigma}) \cdot (1 - p_{\text{base}}), 0, 1)$

TABLE XIX
CATEGORY B: DENSITY-BASED VARIANTS (9 VARIANTS)

ID	Name	Formula
B1	KDE_Scott	$s = -\log \hat{f}_{\text{KDE}}(\mathbf{c})$ — kernel density estimation (Scott bandwidth)
B2	KDE_Sig	$s = \sigma(-0.5 \cdot \log \hat{f}_{\text{KDE}})$ — sigmoid of negative log-density
B3	KDE_Silv	Scott bandwidth $\times 0.5$
B4	KDE_Narrow	Bandwidth $= 0.3\times$ Scott
B5	LOF20	$s = \max(\text{LOF}_{20} - 1, 0)$
B6	GMM_Tanh	$s = \tanh(D_{\text{GMM}}/C)$ — Gaussian Mixture mahalanobis
B7	GMM_Full	$s = -\log P_{\text{GMM}}(\mathbf{c})$ with 3 full-covariance components
B8	GMM_Diag	$s = -\log P_{\text{GMM}}(\mathbf{c})$ with 5 diagonal components
B9	LOF_Corr	Tanh-based correction using LOF signal

6) FP vs. FN Trade-off Analysis:

7) Best Variant for Reducing Missed Fraud: In 10 of 12 combinations, at least one variant achieves **zero missed fraud**. The winners are exclusively Category A (Distance-Based): A4_MahalExp and A1_RobustMahal. However, these achieve 100% recall at the cost of very high FP rates (F1 drops to 0.003–0.048), making them impractical as standalone classifiers but invaluable as **fraud alert generators** in a two-stage pipeline.

8) Summary of V2 Findings: The 66-variant, 792-evaluation experiment yields five principal conclusions:

1) **Quadratic surface fitting (F2) dominates all alternatives.** By learning a degree-2 polynomial correction on the BSDT component vector, F2_QuadSurf achieves Mean F1 $= 0.787$, eliminating 88.1% of false positives:

$$s_{\text{F2}} = \text{clip}\Big(p_{\text{base}} + \phi_2([C, G, A, T])^\top \hat{\boldsymbol{\beta}},\ 0,\ 1\Big) \quad (19)$$

TABLE XX
CATEGORIES C–E: PROJECTION, ISOLATION, AND SVM VARIANTS

ID	Name	Formula
Category C: Projection-Based (7 variants)		
C1	PCA_Recon	$s = \|x - \hat{x}_{\text{PCA}}\|_2$ — PCA reconstruction error (95% variance)
C2	PCA_Mahal	$s = \sqrt{(\mathbf{z} - \boldsymbol{\mu}_z)^\top \Lambda^{-1} (\mathbf{z} - \boldsymbol{\mu}_z)}$
C3	LDA_1D	$s = \lvert w^\top \mathbf{c} - \mu_{\text{proj}} \rvert / \sigma_{\text{proj}}$
C4	LDA_Mahal	Mahalanobis in 1D LDA space
C5	LDA_Proba	$s = P_{\text{LDA}}(y = 1 \mid \mathbf{c})$ — LDA posterior probability
C6	Whitened	$s = \|(\mathbf{c} - \boldsymbol{\mu})/\boldsymbol{\sigma}\|_2$
C7	WhitenPow18	$s = s_{\text{whitened}}^{1.8}$
Category D: Isolation-Based (3 variants)		
D1	IsoForest100	$s = -\text{IF}_{100}(\mathbf{c})$ — 100-tree Isolation Forest
D2	IsoForest10	$s = -\text{IF}_{10}(\mathbf{c})$ — 10-tree light variant
D3	IsoCorr	Clipped tanh correction using Isolation Forest signal
Category E: SVM-Based (5 variants)		
E1	OCSVM_RBF	$s = -f_{\text{SVM}}^{\text{rbf}}(\mathbf{c})$ — One-Class SVM (RBF, $\nu = 0.05$)
E2	OCSVM_Poly	$s = -f_{\text{SVM}}^{\text{poly}}(\mathbf{c})$ — polynomial kernel
E3	OCSVM_Lin	$s = -f_{\text{SVM}}^{\text{lin}}(\mathbf{c})$ — linear kernel ($\nu = 0.1$)
E4	SVM_Corr	Tanh-based SVM correction with gating
E5	SVM_Blend	$s = 0.5\, p_{\text{base}} + 0.5\, \sigma(0.5(s_{\text{SVM}} - \text{med})/\hat{\sigma})$

TABLE XXI
CATEGORIES F–H: QUADRATIC, DISTRIBUTION TRANSFORM, AND INTERACTION VARIANTS

ID	Name	Formula
Category F: Quadratic / Discriminant (5 variants)		
F1	QDA	$s = P_{\text{QDA}}(y = 1 \mid \mathbf{c})$ — Quadratic Discriminant ($\lambda = 0.1$)
F2	QuadSurf	$s = \text{clip}(p_{\text{base}} + \phi_2(\mathbf{c})^\top \hat{\boldsymbol{\beta}}, 0, 1)$ — degree-2 polynomial ridge
F3	DiscrimPM	Quadratic discriminant $\pm\sqrt{\Delta}$
F4	QuadInterR.	Ridge regression on 14 quadratic features
F5	QuadGrid	Grid-search over $s = \text{clip}(p_{\text{base}} + (am^2 + bm)(1 - p_{\text{base}}), 0, 1)$
Category G: Distribution Transforms (7 variants)		
G1	BoxCox	$s = \text{BoxCox}(\text{MFLS} - \min + \epsilon)$
G2	YeoJohnson	$s = \text{YeoJohnson}(\text{MFLS})$
G3	Quantile_U	$s = \frac{1}{4} \sum_j Q_U^{-1}(c_j)$ — quantile-uniform mean
G4	Quantile_G	Quantile-Gaussian $\rightarrow$ Mahalanobis in $\mathcal{N}(0, 1)$ space
G5	RankSigmoid	$s = \sigma(6(\text{rank}(\text{MFLS})/n - 0.5))$
G6	LogitPow	$s = \sigma(-\text{sgn}(l)\lvert l \rvert^{2/5}),\ l = \text{logit}(\text{MFLS})$
G7	YeoJCorr	Tanh-based correction using Yeo-Johnson transformed signal
Category H: Interaction & Enrichment (6 variants)		
H1	Poly2LR	$s = P_{\text{LR}}(y = 1 \mid \phi_2(\mathbf{c}))$ — degree-2 polynomial features
H2	Ratios	$s = P_{\text{LR}}(y = 1 \mid [C/(G + \epsilon), A/(T + \epsilon), CA/(GT + \epsilon)])$
H3	HarmonicMean	$s = 4/(C^{-1} + G^{-1} + A^{-1} + T^{-1})$
H4	GeometricMean	$s = (CGAT)^{1/4}$
H5	LpNorms	$s = P_{\text{LR}}(y = 1 \mid [L_{0.5}, L_1, L_2, L_\infty])$
H6	Entropy	$s = -\sum_j p_j \ln p_j,\ p_j = c_j / \sum_i c_i$

2) **False positives are the primary beneficiary.** Across all top variants, the dominant improvement is FP reduction (-54.9% to -88.1%).
3) **Three variants achieve net FN reduction.** B2_KDE_Sig (-24.3% FN), A9_kNNCorr (-15.9%), and J6_LOF_Pow_Corr (-17.9%) catch strictly more fraud while also reducing false alerts.
4) **LOF+Power chain validates the blind-spot hypothesis.** J6_LOF_Pow_Corr demonstrates that density-based blind spots invisible to the linear MFLS can be exposed through LOF amplification (Shuttle: F1 from 0.134 to 0.938).
5) **Distance-based variants achieve 100% recall** in 10/12 combos, proving that the BSDT component space inherently separates fraud from legitimate transactions. A two-stage pipeline using A4_MahalExp for recall maximisation followed by F2_QuadSurf for precision optimisation would achieve near-perfect fraud detection.

VII. ORTHOGONALITY AND NON-REDUNDANCY PROOF

A. Pairwise Correlation Analysis

Elliptic (active components only — G degenerate):

Eth. Addr. (all four active):

Mean $|r|$ (off-diagonal) on Eth. Addr.: 0.201. The moderate A–T correlation (0.504) reflects a genuine relationship between activity anomaly and temporal novelty, but normalised MI analysis (Section VII-C) confirms they carry largely distinct information.

TABLE XXII
CATEGORIES I–J: HYBRID GATE AND LOF+POWER VARIANTS

ID	Name	Formula
Category I: Hybrid Gates (7 variants)		
I1	Mahal_AND	Mahalanobis $\times$ $(C \cdot G)$ AND gate, tanh correction
I2	KDE_XOR	KDE $\times$ $\tanh(CA - GT)$ XOR gate
I3	Iso_Ratio	IsoForest $\times$ $\tanh(2(r-1))$, $r = (C+G)/(A+T+\epsilon)$
I4	SVM_Tanh	SVM $\times$ $\tanh(\text{sat}) \cdot \sigma(-12(p_{\text{base}} - 0.5))$
I5	LDA_Power	LDA normalised to $[0,1]$, then $s^{1.5}$
I6	EnsVote	Majority vote across A1, B1, D1, E1 at P_{90} thresholds
I7	StackedLR	$s = P_{\text{LR}}(y = 1 \mid [s_{\text{Mahal}}, s_{\text{KDE}}, s_{\text{IF}}, s_{\text{SVM}}])$
Category J: LOF + Power-Law Chain (8 variants)		
J1	LOF_Pow15	$s = (\max(\text{LOF}_{20} - 1, 0) + \epsilon)^{1.5}$
J2	LOF_Pow18	$s = (\max(\text{LOF}_{20} - 1, 0) + \epsilon)^{1.8}$ — primary MFLS-LP
J3	LOF_Pow20	$s = (\max(\text{LOF}_{20} - 1, 0) + \epsilon)^{2.0}$ — aggressive
J4	LOF15_Pow18	Tighter locality ($k = 15$) with power 1.8
J5	LOF30_Pow18	Smoother density ($k = 30$) with power 1.8
J6	LOF_Pow_Corr	$\text{MFLS}' = \text{MFLS} \cdot (\max(\text{LOF}_{20}/\tau, 1))^{1.8}$, tanh correction
J7	LOF_AdaptPow	Adaptive $\gamma = \text{clip}(\lvert\lambda_{\text{YJ}}\rvert + 1, 1.2, 3.0)$
J8	kNN_Pow18	$s = (d_{kNN}/P_{95}(d_{kNN}^{\text{legit}}))^{1.8}$

TABLE XXIII
TOP-10 VARIANTS BY MEAN F1 ACROSS 12 COMBINATIONS

Rank	Variant	Category	Mean F1	ΔF1
1	**F2_QuadSurf**	F	**0.787**	+0.237
2	B2_KDE_Sig	B	0.741	+0.191
3	A9_kNNCorr	A	0.729	+0.179
4	I7_StackedLR	I	0.717	+0.167
5	B6_GMM_Tanh	B	0.713	+0.163
6	I6_EnsVote	I	0.700	+0.149
7	E5_SVM_Blend	E	0.699	+0.149
8	E4_SVM_Corr	E	0.697	+0.146
9	D3_IsoCorr	D	0.688	+0.138
10	B9_LOF_Corr	B	0.681	+0.131

B. Principal Component Analysis

Elliptic (3 active components): PC1 explains 59.8%, PC2 30.4%, PC3 9.8%. All three carry meaningful variance.

Eth. Addr. (4 components): PC1 69.8%, PC2 21.5%, PC3 4.7%, PC4 4.0%. All four contribute.

C. Mutual Information Matrix

Normalised pairwise MI (divided by maximum self-MI):

- **Elliptic:** Mean normalised MI (off-diagonal) $= 0.051$ ✓
- **Eth. Addr.:** Mean normalised MI (off-diagonal) $= 0.064$ ✓

Both are well below the 0.30 threshold for information redundancy, confirming each component captures genuinely distinct information.

D. Variance Inflation Factors

VIF values < 5 indicate acceptable independence.

Eth. Addr.: $C = 1.09$, $G = 1.06$, $A = 1.41$, $T = 1.45$ — all excellent.

Elliptic: $C = 1.30$, $A = 1.17$, $T = 1.24$ — all excellent.

E. Incremental Predictive Contribution

Each component's unique contribution measured by AUC drop when removed from the combined logistic regression:

Temporal Novelty (T) dominates on Elliptic (removing it drops AUC by 15.8 points). On Eth. Addr., the picture is more nuanced: removing A *improves* combined AUC from 0.681 to 0.754, indicating that A introduces noise via its moderate correlation with T ($r = 0.504$) on this dataset. This does not invalidate the framework — rather, it illustrates that the *optimal* component subset is dataset-dependent and that the adaptive weighting mechanism can be extended with feature selection or regularisation. Removing C modestly hurts Elliptic ($\Delta = -0.010$), confirming that not all components contribute positively on every dataset. Proposition 3 holds as stated (combined $>$ best individual), but individual components may exhibit redundancy or interference on specific datasets.

VIII. DISCUSSION

A. Is the Decomposition Complete?

We claim the four components capture the *dominant* failure modes but acknowledge the residual ε. Potential additional components include:

- **Adversarial evasion** — intentionally crafted inputs [23] (distinct from natural camouflage).
- **Label noise** — fraud mislabelled as legitimate in training.
- **Feature interaction blindness** — fraud detectable only through higher-order interactions.

We argue these are either subsumed by existing components or represent Bayes error rather than correctable blind spots.

B. Relationship to the Bias-Variance Decomposition

The classical decomposition [24]:

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise} \tag{20}$$

operates in *error space*. Our decomposition operates in *feature space*:

- C (Camouflage) $\leftrightarrow$ Bias (systematic misclassification)
- T (Temporal Novelty) $\leftrightarrow$ Variance (sensitivity to training sample)

TABLE XXIV
AGGREGATE FRAUD ACCOUNTING (SUMMED ACROSS 12 COMBOS, $N_{\text{TOTAL}} = 77{,}790$ FRAUD INSTANCES)

Variant	Caught	Missed	FP	ΔFP	ΔFP%	ΔFN	ΔFN%	F1	Acc%
BASE	76,046	1,744	44,740	—	—	—	—	0.550	85.9
F2_QuadSurf	75,513	2,277	5,345	−39,395	−88.1	+533	+30.6	**0.787**	**96.3**
B2_KDE_Sig	76,469	1,321	17,368	−27,372	−61.2	−423	−24.3	0.741	85.5
A9_kNNCorr	76,323	1,467	12,878	−31,862	−71.2	−277	−15.9	0.729	93.2
I7_StackedLR	75,192	2,598	7,876	−36,864	−82.4	+854	+49.0	0.717	93.8
B6_GMM_Tanh	75,170	2,620	5,557	−39,183	−87.6	+876	+50.2	0.713	96.3
J6_LOF_Pow_Corr	76,358	1,432	20,191	−24,549	−54.9	−312	−17.9	0.647	83.3

TABLE XXV
SELECTED PER-DATASET FRAUD ACCOUNTING (F2_QUADSURF AND J6_LOF_POW_CORR)

Dataset	Model	Fraud	Method	Caught	Missed	FP	Acc%	F1
Shuttle	XGB93	2,809	Base	2,809	0	36,469	7.15	0.134
			F2_QuadSurf	2,653	156	22	**99.55**	**0.968**
			J6_LOF	2,655	154	198	99.10	0.938
Pendigits	XGB93	125	Base	0	125	0	97.73	0.000
			F2_QuadSurf	117	8	9	**99.69**	**0.932**
			J6_LOF	11	114	12	97.71	0.149
Credit Card	Fresh-XGB	394	Base	284	110	32	99.94	0.800
			F2_QuadSurf	284	110	27	99.94	**0.806**
			J6_LOF	282	112	22	99.94	0.808
Elliptic	Fresh-XGB	33,616	Base	33,426	190	532	98.06	0.989
			F2_QuadSurf	33,419	197	521	98.07	0.989
			J6_LOF	33,394	222	482	**98.11**	**0.990**

TABLE XXVI
FP/FN CHANGE DIRECTION ACROSS 12 COMBOS PER VARIANT

Variant	FP Inc.	FP Dec.	FN Inc.	FN Dec.	Pattern
F2_QuadSurf	7	5	5	6	FP
B2_KDE_Sig	3	5	7	2	**FN**
A9_kNNCorr	3	8	7	4	FN
I7_StackedLR	4	8	5	7	FN
B6_GMM_Tanh	5	4	4	5	FP
D3_IsoCorr	6	4	3	6	FP
J6_LOF_Pow	6	4	4	7	FP

TABLE XXVII
VARIANT MINIMISING MISSED FRAUD PER COMBO

Dataset	Model	Base	Best	New	Red.%
Pendigits	XGB93	125	A1_RobMahal	0	100
Pendigits	Fresh-XGB	21	A1_RobMahal	0	100
Mammography	XGB93	208	A4_MahalExp	0	100
Mammography	Fresh-XGB	80	A4_MahalExp	0	100
Eth. Addr.	XGB93	212	A4_MahalExp	0	100
Eth. Addr.	Fresh-XGB	340	A4_MahalExp	0	100
Shuttle	Fresh-XGB	64	A1_RobMahal	0	100
Credit Card	XGB93	394	A4_MahalExp	0	100
Credit Card	Fresh-XGB	110	A4_MahalExp	0	100
Elliptic	Fresh-XGB	190	A4_MahalExp	0	100

- G (Feature Gap) ↔ Noise (irreducible information loss)
- A (Activity Anomaly) ↔ a novel component not captured classically

TABLE XXVIII
PEARSON CORRELATIONS — ELLIPTIC

	C	A	T
C	1.000	−0.350	−0.414
A	−0.350	1.000	0.279
T	−0.414	0.279	1.000

TABLE XXIX
PEARSON CORRELATIONS — ETH. ADDR.

	C	G	A	T
C	1.000	−0.043	−0.089	−0.276
G	−0.043	1.000	−0.226	−0.068
A	−0.089	−0.226	1.000	0.504
T	−0.276	−0.068	0.504	1.000

TABLE XXX
INCREMENTAL CONTRIBUTION — ELLIPTIC

Dropped	AUC (full)	AUC (red.)	ΔAUC
C	0.923	0.933	−0.010
A	0.923	0.924	−0.001
T	0.923	0.765	**+0.158**

C. Generalisability Beyond Blockchain

The four failure modes are defined using general supervised classification concepts (decision boundaries, feature coverage, distribution shift, activity profiles). Nothing is blockchain-specific. Cross-domain validation (Section VI-H) **empirically confirms** this:

- Credit card fraud — MFLS AUC $= 0.885$, recall $+19.0$pp

TABLE XXXI
INCREMENTAL CONTRIBUTION — ETH. ADDR.

Dropped	AUC (full)	AUC (red.)	ΔAUC
C	0.681	0.670	+0.011
G	0.681	0.694	−0.013
A	0.681	0.754	**−0.073**
T	0.681	0.681	+0.000

- E-commerce CNP fraud — MFLS AUC $= 0.638$, recall $+0.9$pp
- Aerospace telemetry — MFLS AUC $= 0.982$, recall $+28.9$pp
- Medical imaging — MFLS AUC $= 0.916$, recall $+100.0$pp
- Digit recognition — MFLS AUC $= 0.972$, recall $+100.0$pp

Under the strict transfer protocol (Section VI-H), the four BSDT components generalise across all seven datasets, achieving mean ExpGate F1 $= 0.599$ with mean FDR $= 40.1\%$ (Table VI). Per-dataset ExpGate F1 ranges from 0.266 (IEEE-CIS) to 0.953 (Shuttle). The adaptive weighting mechanism automatically discovers which component dominates in each domain: T for blockchain, A for banking, C for aerospace. **The decomposition structure is universal, but the dominant component varies by domain.** IEEE-CIS is the hardest transfer case (calibrated AUC $= 0.655$), yet the consistent Base $\rightarrow$ MFLS $\rightarrow$ SignedLR $\rightarrow$ QuadSurf $\rightarrow$ ExpGate progression is preserved.

We believe the theory extends further to insurance claims fraud, tax evasion detection, and cyber intrusion detection, where the same four failure modes apply.

D. Limitations

1) **Single base model.** *(Addressed in Section VI-L.)* Validated across XGBoost, LightGBM, RandomForest, and Logistic Regression. Future work: TabNet, FT-Transformer, GNNs, and deep autoencoders.
2) **Component interactions.** *(Fully addressed in Sections VI-M and VI-N.)* The 66-variant exploration demonstrates that non-linear interactions are the primary driver of improvement. F2_QuadSurf captures all pairwise and squared interactions, achieving Mean F1 $= 0.787$.
3) **Causality.** The components correlate with missed fraud but causal mechanisms remain unestablished.
4) **Precision trade-off.** On out-of-domain datasets where the base model has zero recall, BSDT correction achieves high recall at reduced precision. As shown in Section VI-J, the signed logistic variant substantially mitigates this trade-off, reducing mean FDR from 86.1% (+MFLS) to 47.3% (SignedLR) under the strict transfer protocol when a small calibration set is available. The V2 experiment (Section VI-N) further demonstrates that **false positives are the primary beneficiary**: F2_QuadSurf eliminates 88.1% of aggregate false positives (from 44,740 to 5,345), and three variants reduce both FP and FN simultaneously.
5) **Statistical power.** The main seven-dataset results use 2 random seeds; only the V2 student-model evaluation uses 5 seeds. A full multi-seed analysis across all datasets is needed to quantify variance. The reported ±std values on the Eth. Addr. V2 evaluation provide a lower bound on stability.
6) **No external baselines.** We compare BSDT correction operators against each other and against the uncorrected base model, but do not benchmark against alternative post-hoc correction methods (e.g., conformal prediction, cost-sensitive re-weighting, selective classification). Direct comparison with such methods is an important direction for future work.
7) **Correction operators require target-domain labels.** The abstract's "without retraining" refers to the frozen base model; SignedLR, QuadSurf, and ExpGate are *fit per dataset* on the train-fit split. This is lightweight supervised adaptation (a logistic regression or ridge regression on 4–15 features), not zero-label transfer.
8) **Anomaly benchmarks $\neq$ fraud.** Four of the seven datasets (Shuttle, Mammography, Pendigits, Credit Card) are anomaly-detection benchmarks repurposed as fraud proxies. While the BSDT components are defined in terms of general classification failure modes, domain-specific fraud dynamics (adversarial adaptation, regulatory context) are absent from these benchmarks.

IX. PRACTICAL IMPLICATIONS

A. For Model Developers

Append the seven supplementary features φ_1–φ_7 to training data before retraining. This allows the model to "see its own blind spots," learning which combinations of camouflage, gap, activity, and novelty indicate hidden fraud.

B. For Compliance Officers

Use MFLS as a **secondary screening score**. Transactions with $\hat{p}(x) < \tau$ but $\mathrm{MFLS}(x) > 0.5$ should be routed to manual review, capturing missed cases without overwhelming the review queue.

C. For Regulators

BSDT provides a **standardised vocabulary** for ML model limitations. Rather than opaque accuracy numbers, regulators can ask: "What is this model's camouflage vulnerability? What is its temporal novelty coverage?"

D. When to Use BSDT vs. Retrain

Use BSDT correction when:
- High-recall priority (AML screening, regulatory compliance)
- Base recall $< 50\%$ and retraining infeasible
- Speed matters — correction is instantaneous, requires no labelled data

Retrain with MFLS features when:
- Base recall $> 70\%$ in-domain
- Precision matters equally or more than recall

TABLE XXXII
V10 SUMMARY VS. BASE MODEL

Metric	Base	V10	Δ
Mean % Correct	85.9%	95.7%	+9.8pp
Mean % Missed	34.8%	21.8%	−13.0pp
Mean F1	0.551	0.740	+0.189
Mean AUC	0.717	0.937	+0.220
OOD % Correct	87.5%	99.0%	+11.5pp
OOD F1	0.425	0.695	+0.270

TABLE XXXIII
F2_QUADSURF (V2) VS. V10 AND BASE

Metric	Base	V10	F2
Mean F1	0.551	0.740	**0.787**
Mean Accuracy	85.9%	95.7%	**96.3%**
Aggregate FP	44,740	—	**5,345**
OOD F1	0.425	0.695	**0.763**

- Labelled in-domain data is available

Use both when deploying a retrained model but wanting a safety net for concept drift.

X. CONCLUSION

We have proposed the Blind Spot Decomposition Theory, a framework that characterises ML fraud detection failures through four measurable, near-orthogonal components. The theory is:

1) **Testable** — any dataset's missed fraud should be dominated by C, G, A, T (plus residual ε).
2) **Falsifiable** — discovery of a dominant fifth mode orthogonal to the existing four would expand the basis.
3) **Useful** — recovery of up to 84.4 percentage points of recall without base-model retraining (correction operators are fit per dataset on a train-fit split).
4) **Adaptive** — self-calibrating weights eliminate manual tuning.
5) **Universal** — validated across 7 datasets, 6 domains, 6 model architectures, and ∼1.16M samples with mean MFLS AUC of 0.857 and mean ExpGate F1 of 0.599.
6) **Model-agnostic** — consistent improvement across 36 model–dataset combinations.

The recommended V10 integration (5-feature logistic regression on $[\hat{p}(x), C, G, A, T]$) outperforms all alternatives:

The 66-variant exploration (Section VI-N) provides exhaustive validation:

The framework transforms missed fraud from an opaque model residual into a structured, interpretable, and correctable phenomenon. Under a strict four-way split protocol that eliminates label leakage (Section VI-H), the four BSDT components lift a frozen transfer model from mean F1 $= 0.153$ to mean ExpGate F1 $= 0.599$ with mean FDR $= 40.1\%$ across seven datasets spanning six domains, with per-dataset ExpGate F1 ranging from 0.266 (IEEE-CIS) to 0.953 (Shuttle). The consistent progression Base → MFLS → SignedLR → QuadSurf → ExpGate holds on all seven datasets, confirming that the decomposition structure generalises without any domain-specific retraining.

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TABLE XXXIV
SYMBOL DEFINITIONS

Symbol	Definition
$x \in \mathbb{R}^d$	Transaction feature vector ($d = 93$)
$y(x) \in \{0, 1\}$	True label (0 = legit., 1 = fraud)
$f(x) = \hat{p}(x)$	Predicted fraud probability
τ	Classification threshold
$\mathcal{M}_f$	Set of missed fraud transactions
$S_i(x)$	i-th component score ($i \in \{C, G, A, T\}$)
w_i	Weight for component i
λ	Correction strength parameter
$\boldsymbol{\alpha}$	Dirichlet concentration parameters

Algorithm 1 Adaptive MFLS Pipeline

Require: Dataset X, model f, [optional: labels y]
Ensure: Corrected predictions p^*

```
Compute reference statistics from X
Compute S = [C(X), G(X), A(X), T(X)]
if labels y available then
    w ← MI_weights(S, y)
else if no labels then
    w ← VR_weights(S, f(X))
else
    w ← Bayesian_update(S, feedback)
end if
Compute MFLS = S · w
Grid search optimal λ and τ
p*(x) = p(x) + λ · MFLS(x) · (1 − p(x)) · 1[p(x) < τ]
return p*
```

APPENDIX A
FULL MATHEMATICAL SPECIFICATION

A. Notation

B. Algorithm: Adaptive MFLS Pipeline

APPENDIX B
INVARIANCE ANALYSIS

We characterise the symmetry and equivariance properties of the four BSDT components and the core MFLS correction operator. Understanding which invariances hold (and which break) serves two purposes: **(i)** it identifies the preprocessing assumptions under which BSDT is valid, and **(ii)** it establishes that MFLS correction is provably safe ($\hat{p}^* \geq \hat{p}$). The analysis distinguishes the *core theory* (components + MFLS) from the *deployment-layer operators* (SignedLR, QuadSurf, ExpGate), which are application choices rather than part of the decomposition itself.

A. Notation

Let $x \in \mathbb{R}^d$ be a feature vector, $S(x) = [C(x), G(x), A(x), T(x)]$ the component vector, and $\hat{p}(x)$ the base model's fraud probability. We write $\hat{p}^*(x)$ for the corrected probability under a given operator. We analyse six core properties at the component level (scale, translation, permutation, affine equivariance, label dependence, dimensionality stability) and two algebraic properties of the MFLS score (monotonicity and correction safety).

B. Component-Level Invariances

Proposition 5 (Component invariance profile)**.** *Under the BSDT component definitions:*

(i) ***Scale.*** *None of C, G, A, T is scale-invariant. G is approximately invariant for $|\alpha| \gg \epsilon$ (threshold effect) but collapses to 1 as $\alpha \to 0$.*

(ii) ***Translation.*** *No component is translation-invariant. C depends on $\|x - \mu_{legit}\|$; G depends on the absolute zero threshold; A and T depend on reference statistics.*

(iii) ***Permutation.*** *G is permutation-invariant (it counts near-zero features regardless of position). C, A, and T are not: C depends on per-feature μ_{legit}; A accesses specific column indices; T uses per-feature reference moments. This is by design — the decomposition deliberately tracks* which *features are anomalous, not merely* how many.

(iv) ***Dimensionality ($d \to \infty$).*** *T converges to $\sigma(-0.5) \approx 0.378$ under the null by the law of large numbers on normalised squared deviations (empirically: std$(T) < 0.011$ at $d = 1000$). G converges to $P(|X_j| < \epsilon)$. A is dimension-independent (uses 2 columns). C drifts toward 0 as d grows because Euclidean distances concentrate; this is mitigated by the 99th-percentile normalisation but not eliminated.*

Proof. (i)–(iii): Direct from the formulae. For (i), $C(\alpha x) = 1 - \|\alpha x - \mu\|_2 / d_{\max} \neq C(x)$ unless $\alpha = 1$. For (iii), $G(\Pi x) = |\{j : |x_{\pi(j)}| < \epsilon\}|/d = |\{j : |x_j| < \epsilon\}|/d = G(x)$. (iv): For T, the normalised mean $\bar{m}(x) = d^{-1} \sum_j (x_j - \bar{x}_j^{\text{ref}})^2 / \text{Var}_j^{\text{ref}}$ converges a.s. by the SLLN when feature deviations are i.i.d. with finite second moment. Empirical verification on synthetic Gaussian data ($d \in \{10, 50, 93, 200, 500, 1000\}$) confirms convergence of T and drift of C (Table XXXV). □

C. MFLS Monotonicity and Correction Safety

The MFLS composite score $\text{MFLS}(x) = \sum_i w_i S_i(x)$ is the central object of the decomposition theory. Two algebraic properties make it suitable as a correction signal.

Proposition 6 (MFLS core properties)**.** *(i)* ***Monotonicity.*** *MFLS is monotone-equivariant: if $S_i(x) \leq S_i(x')$ for all i, then MFLS$(x) \leq$ MFLS(x'). This follows from $w_i \geq 0$ across all weighting schemes (MI, Fisher VR, Bayesian, gradient). Monotonicity ensures that higher blind-spot evidence always produces a higher composite score, making MFLS interpretable as a* severity measure. *Empirical verification: 45 comparable pairs tested, 0 violations.*

(ii) ***Correction safety.*** *The MFLS correction formula $\hat{p}^*_{MFLS} = \hat{p} + \lambda \cdot MFLS \cdot (1 - \hat{p}) \cdot \mathbb{1}[\hat{p} < \tau]$ guarantees $\hat{p}^*(x) \geq \hat{p}(x)$ for all x. Since $\lambda \geq 0$, MFLS $\in [0, 1]$, and $(1 - \hat{p}) \geq 0$, the additive correction term is non-negative. This is a theorem, not an empirical observation:* ***MFLS correction can never reduce the probability of flagging fraud****.*

Proof. (i): $w_i \geq 0$ for all four schemes by construction (MI $= I(S_i; Y) \geq 0$; Fisher VR $= \text{Var}_{\text{between}}/\text{Var}_{\text{within}} \geq 0$; Bayesian posteriors are non-negative; gradient updates maintain $w_i \geq 0$

via projection). Thus $\text{MFLS}(x) = \sum_i w_i S_i(x)$ is a non-negatively weighted sum, and component-wise domination implies sum domination.

(ii): Each factor is non-negative: $\lambda \geq 0$, $\text{MFLS}(x) \in [0,1]$, $(1-\hat{p}(x)) \geq 0$, $\mathbb{1}[\hat{p} < \tau] \in \{0,1\}$. Therefore $\hat{p}^* - \hat{p} = \lambda \cdot \text{MFLS} \cdot (1-\hat{p}) \cdot \mathbb{1}[\hat{p} < \tau] \geq 0$. □

These two properties distinguish MFLS from generic post-hoc correction methods. Calibration (Platt scaling, isotonic regression) can both increase *and* decrease predictions. MFLS is constrained to only *boost* predictions for samples with blind-spot evidence, ensuring it functions as a safety net rather than a general recalibration tool.

D. Operating Conditions

a) Affine equivariance and the re-fitting requirement.: BSDT components are defined in the preprocessed feature space (sign-log + RobustScaler). Under an arbitrary affine transformation $x \mapsto Ax + b$ without re-fitting reference statistics, components change by a mean absolute deviation of up to 0.13 (Camouflage). When reference statistics are re-fit on the transformed data, the deviation drops substantially (to < 0.06 for C, < 0.05 for A, < 0.04 for T; G is exactly invariant).

Corollary 1 (Re-fitting requirement)**.** *BSDT components are not affine-equivariant with fixed reference statistics. Domain adaptation requires re-fitting μ_{legit}, $d_{\max}$, $\bar{x}^{ref}$, and Var^{ref} on the target-domain train-fit split. Only G is invariant without re-fitting. This is the formal justification for the 4-way split protocol (Section VI).*

b) Label dependence.: Three of four components require labelled data for reference statistics (C: legitimate centroid, A: fraud activity stats, T: fraud reference distribution). Only G (Feature Gap) is label-free. Among weighting schemes, Fisher Variance-Ratio weights are unsupervised (using model-output splits only), enabling a fully label-free path: G + VR weights → MFLS. This is not a design accident — it falls out of the invariance analysis as the unique fully unsupervised combination.

c) Dimensionality stability.: Table XXXV summarises the convergence behaviour. The key result is that T is provably stable under the LLN as $d \to \infty$ (std drops from 0.083 at $d = 10$ to 0.011 at $d = 1000$), while C suffers from Euclidean distance concentration (mean drifts from 0.720 to 0.325). For high-dimensional feature spaces ($d > 500$), C should be computed in a reduced subspace or replaced with a cosine-distance variant.

E. Deployment Operators: Algebraic Properties

The deployment-layer operators (SignedLR, QuadSurf, ExpGate) are *application choices* for how to convert the MFLS diagnostic into a corrected probability. They are not part of the core decomposition theory but have distinct algebraic properties that guide operator selection.

TABLE XXXV
DIMENSIONALITY SENSITIVITY OF BSDT COMPONENTS (SYNTHETIC GAUSSIAN, $n = 1000$, 5% FRAUD RATE).

d	$\bar{C} \pm \sigma$	$\bar{G}$	$\bar{T} \pm \sigma$
10	0.720 ± 0.172	0.000	0.416 ± 0.083
50	0.676 ± 0.144	0.000	0.413 ± 0.034
93	0.654 ± 0.135	0.000	0.417 ± 0.026
200	0.565 ± 0.120	0.000	0.421 ± 0.019
500	0.455 ± 0.094	0.000	0.419 ± 0.014
1000	0.325 ± 0.064	0.000	0.414 ± 0.011

TABLE XXXVI
INVARIANCE AND ALGEBRAIC PROPERTIES. LEFT: CORE THEORY (COMPONENTS + MFLS); RIGHT: DEPLOYMENT OPERATORS. ✓ = HOLDS; ✗ = VIOLATED; ≈ = APPROXIMATELY HOLDS; — = NOT APPLICABLE.

Property	C	G	A	T	**MFLS**	SLR	QS	EG
Scale invariance	✗	≈	✗	✗	✗	✗	✗	✗
Translation invariance	✗	✗	✗	✗	✗	✗	✗	✗
Permutation invariance	✗	✓	✗	✗	✗	✗	✗	✗
Monotone equivariance	—	—	—	—	✓	✗	✗	✗
Safety ($\hat{p}^* \geq \hat{p}$)	—	—	—	—	✓	✓	✗	✗
Label-free	✗	✓	✗	✗	VR	✗	✗	✗
Idempotent	—	—	—	—	✗	✓	—	—
Dim-stable ($d \to \infty$)	✗†	✓	✓	✓	✓	✓	✗‡	✗‡

† C drifts toward 0 due to distance concentration. ‡ Polynomial feature count grows as C

a) SignedLR.: Uses logistic regression on the four components with a $\max(\hat{p}, p_{\text{SLR}})$ formulation. It is correction-safe by construction and idempotent ($\max(a, \max(a,b)) = \max(a,b)$), but *not* monotone: learned coefficients β_i can be negative (empirically, $\beta_C = -1.92$, $\beta_T = -7.41$). The negative β_C is practically important — it means camouflage *reverses direction* in cross-domain transfer. A transaction that looks "camouflaged" in the source domain is less likely fraud in the target domain. This explains why naive MFLS (which treats C as always positive evidence) has high FDR, while SignedLR learns to correct for it.

b) QuadSurf and ExpGate.: Fit polynomial or gated residual surfaces. They are *bidirectional* corrections: $\hat{p}_{\text{quad}} = \text{clip}(\hat{p} + \boldsymbol{\beta}^\top \phi_2(S), 0, 1)$, where $\boldsymbol{\beta}^\top \phi_2(S)$ is unconstrained in sign. Empirically, 73.2% of samples receive a *lower* corrected probability. This is not a defect — it is the mechanism by which QuadSurf/ExpGate suppress false alarms, reducing FDR from 46.9% (SignedLR) to 36.0% (QuadSurf). They trade the safety guarantee for expressiveness (bidirectional probability adjustment).

c) Operator composition.: The chain is non-commutative: SignedLR targets classification, QuadSurf targets residual regression, and reversing their order changes the objective surface.

F. Invariance Summary

Table XXXVI consolidates the analysis. The central result is that MFLS — the core contribution of this paper — satisfies both monotonicity and correction safety, properties that no deployment-layer operator fully preserves. SignedLR

preserves safety but sacrifices monotonicity to gain expressiveness. QuadSurf and ExpGate sacrifice both but gain bidirectional correction capability, which is empirically valuable for FDR reduction. The choice of deployment operator is thus a *safety–expressiveness trade-off* that is orthogonal to the validity of the underlying decomposition.

APPENDIX C
CHOICE OF FUNCTIONAL FORMS AND SENSITIVITY

Each component formula satisfies three design constraints: **(i)** output in $[0, 1]$, **(ii)** monotonicity in failure-mode severity, and **(iii)** computability from the feature vector alone (no model internals required).

A. *Camouflage* — $C(x) = 1 - \|x - \mu_{legit}\|_2 / d_{\max}$

Measures geometric proximity to the legitimate cluster. The 99th-percentile cap prevents outlier distortion. In sensitivity tests on Elliptic, replacing Euclidean with cosine distance changed C scores by < 0.04 mean absolute deviation and left MFLS AUC within ± 0.01.

B. *Feature Gap* — $G(x) = |\{j : |x_j| < \epsilon\}|/d$

A counting measure — what fraction of features carry no information. The threshold $\epsilon = 10^{-8}$ distinguishes true zeros from floating-point noise. Robust because G is either active (sparse data) or degenerate (dense data).

C. *Activity Anomaly* — $A(x)$

The $\log(1 + |\cdot|)$ transform stabilises variance of heavy-tailed transaction counts. Z-scoring against the caught-fraud distribution measures deviation from what the model has learned. Replacing $\sigma(\cdot)$ with min-max normalisation changed MFLS AUC by < 0.008 on both blockchain datasets.

D. *Temporal Novelty* — $T(x)$

A soft normalised Mahalanobis-type distance. The shift by -2 and scale by $1/2$ centres the sigmoid transition at moderate novelty ($\bar{m} \approx 2$). Varying the centring constant from 1.5 to 3.0 shifted MFLS AUC by < 0.015.

E. *General Sensitivity Summary*

The MFLS framework is intentionally robust to functional-form choices because adaptive weight calibration absorbs much of the sensitivity. A component with a sub-optimal formula receives a slightly different weight but produces a similar combined MFLS. The framework's power arises from the *decomposition structure* (four orthogonal failure axes), not individual component precision.